After Ford

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The moment of Detroit's deepest crisis coincides with the 'Death of Modern Architecture' as announced by Charles Jencks in 1977. (1) This is no coincidence. The emergence of postmodern architecture and urbanism in the seventies – sweeping the market in the eighties – represents much more than a new aesthetic sensibility. The postmodern rejection of homogeneity, coherence, and completeness; and the explicit celebration of heterogeneity mark a radical departure from fifty years of modernist development. The force behind these developments, rather than emerging from within the architectural discipline itself, is to be be found on the socio-economic level. Postmodern cultural production coincides with the historical crisis in the regime of mechanical mass-production, first developed by Ford in Detroit. (2)

The historical closure of fordism as a model of socio-economic progress spelled the demise of Detroit, once the proud origin of modern industrial development. "Detroitism" became a globally emulated recipe for economic prosperity. Now Detroit stands devastated; overburdened by the infrastructural, architectural and human sediment of its fordist past. Central parts of Detroit are empty, large buildings stand as ruins: offices, schools, train stations and vast urban territories have been abandoned. Urban planning proposals counter this drastic situation with equally drastic measures: The demolition of whole urban quarters and their conversion into parks. Greenbelts are proposed to cut the vast, fragmented field into recognizable "communities", sealing the ultimate fate of Detroit: to become the suburb of its own suburbs. Those extended suburbs are alive and well, forming a polycentric conurbation where typically post-fordist service industries settle at a safe distance from inner city wastelands.

But it would be wrong to assume that post-fordism is the era of suburbia and fordism the era of the city. Suburbanization was the general rule of (mature) fordist urbanization. Postfordism breaks the universality of suburbanization. The new model of post-fordist urbanism re-inhabited the historic city. Postmodern architecture found its market in the rediscovery and "detournement"(3) of the historical city not merely as brandable commodity but as necessary communication hub for the new economy. Jane Jacobs rendered a critical verdict on Detroit in 1961, at the height of its economic power. "Virtually all of Detroit is as weak on vitality and diversity as the Bronx. It is ring superimposed upon ring of failed gray belts. Even Detroit¹s downtown itself cannot produce a respectable amount of diversity. It is dispirited and dull, and almost deserted by seven o'clock of an evening." (4) Monotony, lack of diversity; these are the typical "ills" or "failures" of the modern city. To avoid Jacob¹s ahistorical condemnation of the industrial city, one must grasp the economic rationality underpinning its development. This includes the intentional rationality and social meaning of urban monotony, zoning, and the various symptoms of industrialized urban arrangement. Over half a century of rationally planned coherent city building can not have been a "mistake". But what was progressive then has indeed become dysfunctional today. The new socio-economic logic of Postfordism offers a reading of the current prospects of Detroit and other cities caught in the dynamic of global economic restructuring. Any understanding of Detroit must begin with the socio-economic logic of fordism and its urban implications.

Fordism as a Technical and Spatial System

Detroit served as a visible model of fordist industrial development during the first half of the twentieth century. As an economic monoculture it mirrored the prosperity, growth and decline of the automobile industry. Detroit offers a paradigmatic case study of fordism as an organizational model of urbanization and for the collusion between industry and architecture, as personified by the collaboration between Henry Ford and Albert Kahn. One might speak of three phases of the fordist revolution: (5)

Phase 1: Taylorization takes command.

Automobile manufacturing in the pioneering days is organized around the work of autonomous artisan-engineers. To increase the speed and scale of production, Ford applies Taylor¹s principles of scientific management. Work becomes a scientific object, observable, controllable and modifiable. Individual laborer¹s tasks are recorded, analyzed and broken down into elementary movements. Efficiency is optimized by the reconfiguration of tasks within time and space according to the dialectic of differentiation and repetition. Within this concept of order the flow of production over time is the controlling parameter. Albert Kahn provides the required architecture and spatial organization. The Kahn System of Reinforced Concrete enables wide spaces offering freedom of movement and flexibility for functional adaptation to various production lines. Ford¹s Highland Park plant (1909) with large expanses of clear space allowed the unconstrained organization of various production cycles, each on its floor. Discrete processes were stacked vertically, joined via floor openings and fed by a flow of material from top to bottom. This vertical architectural organization enabled the production of the first complex assembly-line product: Ford's Model T. (6)

Phase 2: The factory under one roof is super-ceded.
The assembly line concept is applied to an overall urban complex. Several single story buildings are joined together, each accommodating a specific task, and extruded to the length desired. Entire buildings act as elements of multi-building assembly lines. At the River Rouge plant (begun 1917) the flow of materials and sub-components determine the overall "urban" layout as an integrated machine. This is literally the "city as machine" later proclaimed by the ideologues of modernist urbanism (Le Corbusier, Hilberseimer etc.).

Phase 3: Production patterns are decentralized.
Just after creating the world's largest industrial complex at River Rouge, Ford proposed a decentralizing anti-urbanism. Fordist decentralization involves the re-application of fordist principles of production on a regional and national scale. Kahn's task became the construction of specialized production sites scattered across the country, and linked by infrastructural networks. This phase of fordism shaped a regional and then nation-wide division of labor and demanded the subsequent construction of national highway and communication systems. This extension of fordist productive patterns fueled the rapid decompression of urban industrial cities and the decentralization of both mass production and mass consumption.

The economic success of fordist principles in the US found a multitude of echo-effects abroad. Fordism's controllable mode of production and consumption (with equally calculable profits) found its own interpretation and implementation through various political systems internationally. Fordist production techniques were sought after and implemented in Nazi-Germany as well as in the Soviet Union of Stalin, where Albert Kahn realized 500 major production complexes (1929-1932) under the developmental program of the famous first Five Year Plan.

Fordism as a System of Total Social Reproduction

Fordism is a generalization of the production principles and policies of the Ford Corporation. "Mr. Ford is not a human creature. He is a principle, or better, a relentless process."(7) "Fordism" in 1920's Europe signified the possibility of social progress through new forms of comprehensive industrialization. Ford made social advance tangible through high, universal wages (the famous five dollars per day) while allowing for an eight-hour day and a forty-hour work week three decades before these norms were legislated by the Fair Labor Standards Act of

1938. Fordism gave workers access to the results of their productive efforts as the new scales of mass production turned luxury goods like the motor car into achievable commodities for every worker. The system could reproduce its own market in a self-fulfilling prophecy of economic expansion. The material basis of modern mass society and the “American dream” was established.

As a production system fordism is premised on Taylorism, i.e. the scientific decomposition of the work process into a system of measurable operations. This required the transference of production knowledge from the worker into the mechanism of the assembly line. At each point only the most basic, repetitive task is required, leading to the homogenization of individual labor. This endless repetition and mindlessness of daily labor afforded access to the consumption of universal mass products. Fordism understood as a socio-economic category, rather than a merely technological paradigm, presupposes the systematic integration of the reproduction of labor into a new and totalizing capitalist cycle. The advance of fordism (8) was a qualitative shift in the ability of industry to render workers[1] “basic needs” (food, clothes, shelter, transport, etc.) the object of comprehensive commodification.

The Fordist Logic of Modern Architecture and Urbanism

The totalizing notion of fordism became instrumental to the underlying rationality of modern architecture and urbanism. In Europe this regime of fordist urbanization became possible after the working class (through the mediation of social democracy) gained a degree of power sharing after World War I, establishing the socio-economic basis for modern architecture. (11) These developments implied a revolution in the leadership of the architectural profession. The academic stylists of the imperial institutions were replaced by self-educated architects (Behrens, Gropius, Corb, Mies) who re-invented the discipline by identifying in the mundane (mass housing, mass produced domestic furnishings, factories) worthy and urgent tasks for a modern architecture. The social democratic institutions of the welfare state became the mechanisms through which modern urbanism was advanced. The Fordist task posed was the development of optimally efficient standards and the taylorization of modern living. The house for the “Existenzminimum” became the universal receptacle for a series of universal mass consumer goods: living room, dining set, (Frankfurt-) kitchen, bathroom, washing machine, and later the refrigerator, television and automobile. The new paradigm of Functionalism implied an objectification and analysis of the design process and architectural composition was assimilated to the principles of fordist organization: decomposition, differentiation, repetition and integration.

This logic is evident in the organization of separate functions into specialized and separately optimized volumes. The Dessau Bauhaus is paradigmatic in this respect where residential, administrative and workshop functions are separately articulated, allowing for depth, height and facade to be independently determined for each respective function. The same principles are at work in the canonical conception of the modernist city. Le Corbusier's Ville Radieuse (1933) is the most comprehensive and rigorous application of this logic of differentiation (zoning and distinct functionalist articulation of each zone), repetition (homogeneity of each zone) and hierarchical integration (transport system). Lafayette Park (1955) by Mies, Hilberseimer, and Caldwell offers the most legible post-war example of these principles of modernist planning applied to the renovation of the city of Detroit. Hilberseimer's 1949 publication The New Regional Pattern rendered these same fordist principles of decentralization and differentiation, by intertwining transportation, communication, and production infrastructures across the natural environment of North America.

From Fordism to Postfordism

In the late sixties the fordist system of universal mass production, corporate concentration, collective bargaining and state-regulation was challenged on all fronts. The first serious break in the post-war boom occurred with the recession of 1966/67. The political struggles of 1968, the oil-crisis in 1973, the breakdown of the international exchange-rate system, and a deepening of the recession in 1974 followed. The automobile industry was in free-fall and Detroit, site of the oldest and least competitive plants, was hit hardest. By the end of the seventies it was clear that the recession had become a structural (systemic) crisis that called for new political and economic strategies (12).

The origins of the crisis in fordism and an outline of emergent postfordist tendencies can be found in several concurrent socio-economic transformations. Among these were shifting commodity markets, increasing electronic control of production, decreasing state regulation, increasingly global capital markets, and deteriorating labor relations. Market Stratification: With the growing complexity of the division of labor and the proliferation of white-collar labor, salary stratification increased. Affluence beyond the saturation of the most basic needs meant that markets began to diversify, allowing for status and identity consumption to accelerate aesthetically motivated product-cycles. These developments placed a reward on innovation and flexibility rather than simple cost reduction achieved through mass-market economies of scale. The house, as the main site of consumption, was itself drawn into the logic of differential identity, status, and income. The Modernist housing standard ("Existenzminimum") became the very standard against which market differentiation was measured. Postmodernist design, architecture and urbanism catered to this demand and reconceived of the "failed" modern city as a site for destination recreation and brandable post-urban tourism.

Flexible Production: New computer-based production technologies made possible greater product diversity (small runs) without the enormous cost of handicraft production that had previously limited deviations from the standard. The crucial material factor was the micro-electronic revolution that offered greater productivity through desired economies of scope, rather than scale. Flexible specialization became a technological possibility, and the subsequent fluidity of production demanded the dissolution of static fordist labor and management arrangements.

Vanishing State-Regulation: As products and markets differentiate, economies of scale are recuperated through international expansion. The resultant international economic interdependency has the effect of eroding the economic competence of the nation state, and its ability to smooth out disturbances in the business cycle. As markets globalized, the less economically feasible it became to protect national producers. With the increasing internationalization of mobile capital, a withdrawal from Keynesian macro-economic regulation and a systematic dismantling of the social welfare state became inevitable. This process continues to this day, and Detroit serves as one of the most thoroughly developed models of this tendency.

Globalization of Capital Markets: Globalization emerges as a new model of international integration between production and consumption. Increasingly volatility in capital markets results from speculation in "emerging" economies. Outsourced labor and off-shore production optimize profits by driving down wages through international competition. Globalization takes the form of a re-emergence of inter-imperialist rivalries, militarism, enforced austerity programs, the break up of national welfare programs, and a downward pressure on labor-costs. The majority's standard of living, even in the most advanced economies, stagnates or declines while the class disparity increases. (15)

Exploding Labor Relations: The increasing volatility of global markets and the abdication of state responsibility erodes collective bargaining. Capital-labor compromises and state sanctioned collective bargaining agreements are displaced in favor of "free market" neo-liberalism (Reaganomics and Thatcherism.) Downsizing and outsourcing labor becomes the norm, replacing regular employment with increasingly flexible arrangements. This in turn makes markets even more unpredictable. Employment contracts become shorter.

Mobility increases. “Casual labor” and “self-employment” replace regular employment.

Patterns of Postfordist Production

The crisis in fordist production forced a reorganization of corporate structures as they faced a new pace of change and the increasingly global competition for markets. The ongoing organizational revolution tends to render corporate organization non-hierarchical and replaces command and control mechanisms with participatory and open structures. Although the drive of corporate restructuring towards discursive co-operation remains compromised by the systemic barrier of capitalism that hinges authority upon property rather than discourse, the thrust of development tears and shakes the corporate edifice of fordism.

The Space of Corporate Re-organization

The ‘architecture’ of business organization is liquefying. Fordist strategies of rationalization and hierarchy are giving way in favor of post-modern production patterns. These patterns of arrangement reflect not only a response to the economic and material conditions of production, but also portend an equally important transformation in the structure and organization of corporate space itself.

Fordist principles of corporate organization were generalized from their origin in industrial production to the organization of the service sector and ultimately served as a model of state administration. The whole of society was eventually subsumed within this rigid pattern of hierarchical organization. Everywhere a comprehensive, bureaucratic, functional hierarchy allocated rigid job-descriptions and repetitive tasks within coherent chains of command. The modernist pattern of urbanization is the projection of this total social machine into space.

With the failure of stable cycles of reproduction and expansion, post-fordist production paradigms are increasingly organized around principles of decentralization, horizontality, transparency, fluidity, and rapid mutability. Concurrently, the organization and management of these post-fordist processes and other forms of social arrangement are increasingly based on a set of similar post-modern principles. (14) The new tendencies evident in corporate restructuring can ultimately be summarized as follows:

1 flattening of hierarchies into horizontal fields
2 decentralization and devolution of authority/responsibility
3 self-organization rather than bureaucratic task allocation

4 collegial communication and evaluation rather than command and control
5 dispersal and sharing of information and/or technologies
6 team-work, informal or temporary alliances, ‘loosely coupled networks’
7 hybrid conglomerates and ad-hoc assemblages replace integrated entities
8 increasing reliance on outsourcing, temporary and self-employment
9 mutability, mobility, and indeterminacy as positive values
10 processes analogous to ecological or biological systems (15)

These organizational tendencies are presently evolving in response to the challenge of permanent innovation in production. One could expect (and can find emergent in contemporary work) an analogous set of developments in the cultural sphere including the spatialization of these ideas in the making of architecture and urbanism. The possibilities of a post-fordist urbanism are among the many interesting questions raised by Detroit in general and this anthology in particular.

The radical organizational paradigms elaborated by Deleuze & Guattari in the late seventies seem to foreshadow the paradigms of today[1]s corporate restructuring. The arborescent command pyramid of fordist arrangement is mutating towards the rhizomatic plateau upon which leadership (and all other social functions) is distributed in a permanently shifting multiplicity. Functions and positions reveal their mutual dependency within the unlocked dialectic of co-evolution. Permanent transition implies ambiguity and virtuality as new qualities demanded of both people and places. Every point bears the latency of various crossing trajectories. The modernist city with its strictly coded stereotypes and neat allocation of zones – a place for everything and everything at its place – can not serve as the catalyst for these vital processes of networking and self-organization. Cities such as Detroit are abandoned to an entropic demise under the weight of the previous regime, and await an indeterminate future.

Post-Urbanism

As for developments in the spatialization of postfordist principles, the work of the so-called “LA School” cultural geographers and Ed Soja in particular have offered extensive analysis of the coming post-fordist urbanism. Soja’s exploration of postmodern urbanization focuses on metropolitan region of Los Angeles. In as much as LA is one of the world[1]s leading “superprofitable growth poles” it allows us to identify the future of postfordist urbanization. LA in this regard plays the role Detroit once occupied as the “most thoroughly modern (fordist) city in the world”.

Soja’s analysis of LA suggests that contemporary post-fordist patterns of urbanization function as a “mesocosm” that reproduces within its own spatiality the complexity and contradictions of the global economy. “Seemingly paradoxical but functionally interdependent juxtapositions are the epitomizing features.... One can find in Los Angeles not only the high technology industrial complexes of the Silicon Valley and the erratic sunbelt economy of Houston, but also the far-reaching industrial decline and bankrupt urban neighborhoods of rust-belted Detroit or Cleveland. There is a Boston in Los Angeles, a lower Manhattan and a South Bronx, a Sao Paulo and a Singapore." (16)

The simultaneity of growth and decline, locating leading high tech industrial sectors next to abandoned industrial wastelands, and a growing low-wage economy of industrial sweatshops, posits an uphill battle for social control and exacerbates the friction of distance in the "spread city". (17) Soja[1]s postmodern geography (spread city) differs markedly from the process of post-war suburbanization. It is best described as "an amorphous regional complex that confounds traditional definitions of both city and suburb." (18) This postfordist landscape integrates a loose and open network of research, production and service systems, interspersed with leisure environments and alternating expensive residential developments with enclaves of cheap labor. The interpenetration of different activities succeeds even despite the problems of social control and the cost of policing caused by the proximity of populations increasingly polarized along lines of class, race, and ethnicity.

Another marked spatial phenomenon has been superimposed on the polycentric spatiality of the (LA) postfordist landscape that is also evident in Detroit: the decisive re-colonization of corporate headquarters within the downtown core, reversing the trend of the fordist era. This revival of the central business district and selective gentrification of the inner city, including recreational and pseudo-historic tourist events catering to a largely suburban population reflects the postfordist organizational shift in corporate structure along lines of contemporary production and consumption patterns. The ongoing annexation of Detroit by its own suburbs continues apace as suburban (fordist) wealth simultaneously speculates on property values at the both the agricultural perimeter and abandoned industrial center of what remains one of the largest and most prosperous metropolitan regions in the US.

Detroit[1]s precipitous and public demise may have stepped over a kind of critical threshold, offering a uniquely clear-sighted and unequivocal image of post-fordist dis-investment. In this sense, Detroit offers the most legible indictment of fordist patterns of urbanization. The recent and by now regular injections of recuperative capital, evident in the Renaissance Center project, new casinos, sports stadia and other urban "cures", have failed to promote a revitalization of Detroit[1]s downtown. Some already find delight in the ruins, indulging in a voyeuristic aestheticism.

Others are determined to save the city through social missionary work; others hope to spin it, using media hype and political spin doctoring to influence property values through real-estate speculation. Post-fordist analysis of Detroit offers an image of a post-industrial ex-urban center annexed by its own suburbs creating an extensive and non-hierarchical horizontal field of post-urbanization. Will Detroit benefit from this new form of development, and what are the possibilities for practicing urbanism in this context? Will Detroit[1]s already evident future come to pass as a destination tourist commodity and name brandable recreation center engulfed by pockets of abandonment, disinvestment, and decay? If so, even this unenviable future will need to overcome a century of rusty prejudices.

END

Notes:

1. Jencks, Charles, The Language of Post-Modern Architecture, London 1977.
2.Post-fordism as a category of socio-economic periodization is of Marxist provenance and has been the central term of a wide and fruitful debate. See: Ash Amin, p.1, Introduction to "Post-Fordism: A Reader", Oxford / Cambridge MA.
Robin Murray, Fordism and Postfordism, in S. Hall & M.Jacques, New Times, London 1989
David Harvey, The Condition of Postmodernity, Oxford / Cambridge MA. 1989.

3. "All the elements of the cultural past must be "reinvested" or disappear." Asger Jorn, 'Detourned Painting', quoted in Guy Debord's 'Detournement as negation and prelude', Internationale Situationniste #3, December 1959, translated in: Situationist International – Anthology, Knabb, K.(Ed.), Berkeley 1981.

4. Jane Jacobs, The Death and Life of Great American Cities, first published Random House 1961, Pelican Books, Middlesex 1965, p.1962

5. Bucci, Frederico, Albert Kahn, Princeton 1993

6. Ford, Henry, Mass Production, in Encyclopedia Britannica, vol.15, London-New York 1929, p.40, quoted after Bucci, Federico, Albert Kahn, Princeton 1993, p.42. For Albert Kahn[1]s description of the division of labor in architectural production see A.Kahn, Architectural Trend, in Journal of the Maryland Academy of Sciences, vol.II, no.2, April 1931, p.133, quoted after Bucci, Federico, Albert Kahn, Princeton 1993, p.126/127.

7. Josephson, M., Henry Ford, in: Broom, Oc. 5, 1923. Quoted after: Fehl, Gerhard, Welcher Fordismus?, In: Zukunft aus Amerika, Fordismus in der

Zwischenkriegszeit, Stiftung Bauhaus, Dessau 1995

8. The systematic, theoretical (Marxist) notion of fordism (and neo-fordism) was developed by the French Regulation School of economic analysis, initiated by Michel Aglietta. See Aglietta, Michel, A Theory of Capitalist Regulation: The US Experience, London 1979.

9. Ford anticipated this logic and instituted his own corporate welfare scheme for his workers, including quasi-public facilities like hospitals and schools as well as a reliable pension scheme.

10. Marx distinguishes between technical and social divisions of labor (Capital vol. I.) The former refers to the partition and distribution of tasks between operatives within a firm and the latter describes the division of labor between firms integrated only through the market.

11. In Europe this could only be achieved via the social revolutions that tore down 19th century class-societies and established the working masses and their representatives as an organized political force. By demanding participation in the results of industrial productivity, the laboring classes constituted themselves, for the first time, as the primary market for industrial consumer products and as a client for architecture.

12. See UNIDO (United Nations Industrial Development Organization), Structural Change in Industry, Vienna 1979, and OECD (Organization of Economic Co-operation and Development), Positive Adjustment policies: Managing Structural Change, Paris 1983.

13. Overall productivity suffers as long as the world allocation of material and labor resources remains driven by an irrational, militarily guaranteed, and thus ultimately very costly “cheapness” of labor, which allows the squandering of millions of potentially much more productive lives.

14. See (among others):
Cannon, T.: Welcome to the Revolution – Managing Paradox in the 21st Century, London 1996.
Ray, M. & Rinzler,A.: The New Paradigm for Business, L.A. 1993. Peters,T. : Liberation Management: Necessary Disorganisation for Nanosecond Nineties, N.Y. 1993.
Peters, T.: Thriving on Chaos, N.Y. 1987.
Bergquist,W.: The Postmodern Organization – Mastering the Art of Irreversible Change, New York 1993.
Kilduff,m.: Deconstructing Organisations, Academy of Management Review #18.

Blanchard,K.& Johnson,S.: The One Minute Manager, New York 1982.
Bower,J.L.: Disruptive Technologies: Catching the Wave, Harvard Business Review, Jan/Feb 1995.

15. Castells, M. & Hall, P., Technopoles of the World, London & N.Y. 1994.

16. Edward W. Soja, Postmodern Geographies, London, N.Y. 1989.

17. With this internalization of the periphery comes the largest homeless population, soaring rates of violent crime and the largest prison population within the US. The militarization of the world economy finds itself replicated here in the rule of a militarized LAPD. The anti-racist explosion of 1992 testifies to this.

18. Edward W. Soja, Postmodern Geographies, London, N.Y. 1989, p.212.

end.

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